

Scheduling Algorithm

1. FCFS Scheduling

```
GNU nano 8.7.1
#include <stdio.h>

int main()
{
    int n, i;
    int bt[20], wt[20], tat[20];

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter burst time for each process:\n");

    for (i = 0; i < n; i++)
    {
        printf("P%d: ", i + 1);
        scanf("%d", &bt[i]);
    }

    wt[0] = 0;

    for (i = 1; i < n; i++)
    {
        wt[i] = wt[i - 1] + bt[i - 1];
    }

    for (i = 0; i < n; i++)
    {
        tat[i] = wt[i] + bt[i];
    }

    printf("\nProcess\tBT\tWT\tTAT\n");

    for (i = 0; i < n; i++)
    {
        printf("P%d\t%d\t%d\t%d\n",
            i + 1, bt[i], wt[i], tat[i]);
    }

    return 0;
}
```

```
hemasai@Hemasai:~$ cd ~/OSSP_Assignment
hemasai@Hemasai:~/OSSP_Assignment$ mkdir -p Scheduling
hemasai@Hemasai:~/OSSP_Assignment$ cd Scheduling
hemasai@Hemasai:~/OSSP_Assignment/Scheduling$ nano fcfs.c
hemasai@Hemasai:~/OSSP_Assignment/Scheduling$ gcc fcfs.c -o fcfs
hemasai@Hemasai:~/OSSP_Assignment/Scheduling$ ./fcfs
Enter number of processes: 2
Enter burst time for each process:
P1: 2
P2: 34

Process BT      WT      TAT
P1      2       0       2
P2     34       2     36
hemasai@Hemasai:~/OSSP_Assignment/Scheduling$ |
```

2. SJF — Non-preemptive

```

GNU nano 8.7.1
#include <stdio.h>

int main()
{
    int n, i, j;
    int bt[20], wt[20], tat[20];
    int temp;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter burst time:\n");

    for (i = 0; i < n; i++)
    {
        printf("P%d: ", i + 1);
        scanf("%d", &bt[i]);
    }

    for (i = 0; i < n - 1; i++)
    {
        for (j = i + 1; j < n; j++)
        {
            if (bt[i] > bt[j])
            {
                temp = bt[i];
                bt[i] = bt[j];
                bt[j] = temp;
            }
        }
    }

    wt[0] = 0;

    for (i = 1; i < n; i++)
    {
        wt[i] = wt[i - 1] + bt[i - 1];
    }

    for (i = 0; i < n; i++)
    {
        tat[i] = wt[i] + bt[i];
    }

    printf("\nProcess\tBT\tWT\tTAT\n");

    for (i = 0; i < n; i++)
    {
        printf("P%d\t%d\t%d\t%d\n",
            i + 1, bt[i], wt[i], tat[i]);
    }

    return 0;
}

```

```

hemasai@Hemasai:~/OSSP_Assignment/Scheduling$ nano sjf.c
hemasai@Hemasai:~/OSSP_Assignment/Scheduling$ gcc sjf.c -o sjf
hemasai@Hemasai:~/OSSP_Assignment/Scheduling$ ./sjf
Enter number of processes: 3
Enter burst time:
P1: 23
P2: 45
P3: 65

Process BT      WT      TAT
P1      23      0       23
P2      45      23      68
P3      65      68     133
hemasai@Hemasai:~/OSSP_Assignment/Scheduling$ |

```

3. Priority Scheduling

```

GNU nano 8.7.1
#include <stdio.h>

int main()
{
    int n, i, j;
    int bt[20], priority[20], wt[20], tat[20];
    int temp;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter burst time and priority:\n");

    for (i = 0; i < n; i++)
    {
        printf("P%d Burst Time: ", i + 1);
        scanf("%d", &bt[i]);

        printf("P%d Priority: ", i + 1);
        scanf("%d", &priority[i]);
    }

    for (i = 0; i < n - 1; i++)
    {
        for (j = i + 1; j < n; j++)
        {
            if (priority[i] > priority[j])
            {
                temp = priority[i];
                priority[i] = priority[j];
                priority[j] = temp;

                temp = bt[i];
                bt[i] = bt[j];
                bt[j] = temp;
            }
        }
    }

    wt[0] = 0;

    for (i = 1; i < n; i++)
    {
        wt[i] = wt[i - 1] + bt[i - 1];
    }

    for (i = 0; i < n; i++)
    {
        tat[i] = wt[i] + bt[i];
    }

    printf("\nProcess\tBT\tPriority\tWT\tTAT\n");

    for (i = 0; i < n; i++)
    {
        printf("P%d\t%d\t%d\t%d\t%d\n",
            i + 1,
            bt[i],
            priority[i],
            wt[i],
            tat[i]);
    }

    return 0;
}

```

```

P3      65      68      133
hemasai@Hemasai:~/OSSP_Assignment/Scheduling$ nano priority.c
hemasai@Hemasai:~/OSSP_Assignment/Scheduling$ gcc priority.c -o priority
hemasai@Hemasai:~/OSSP_Assignment/Scheduling$ ./priority
Enter number of processes: 2
Enter burst time and priority:
P1 Burst Time: 34
P1 Priority: 56
P2 Burst Time: 2
P2 Priority: 45

Process BT      Priority      WT      TAT
P1      2      45      0      2
P2      34      56      2      36
hemasai@Hemasai:~/OSSP_Assignment/Scheduling$ |

```

4. Round Robin

```

GNU nano 8.7.1
#include <stdio.h>

int main()
{
    int n, i;
    int bt[20], rem_bt[20];
    int wt[20], tat[20];
    int quantum;
    int time = 0;
    int done;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter burst time:\n");

    for (i = 0; i < n; i++)
    {
        printf("P%d: ", i + 1);
        scanf("%d", &bt[i]);

        rem_bt[i] = bt[i];
        wt[i] = 0;
    }

    printf("Enter time quantum: ");
    scanf("%d", &quantum);

    while (1)
    {
        done = 1;

        for (i = 0; i < n; i++)
        {
            if (rem_bt[i] > 0)
            {
                done = 0;

                if (rem_bt[i] > quantum)
                {
                    time = time + quantum;
                    rem_bt[i] = rem_bt[i] - quantum;
                }
                else
                {
                    time = time + rem_bt[i];
                    wt[i] = time - bt[i];
                    rem_bt[i] = 0;
                }
            }
        }

        if (done == 1)
        {
            break;
        }
    }

    for (i = 0; i < n; i++)
    {
        tat[i] = wt[i] + bt[i];
    }

    printf("\nProcess\tBT\tWT\tTAT\n");

    for (i = 0; i < n; i++)
    {
        printf("P%d\t%d\t%d\t%d\n",
            i + 1,
            bt[i],
            wt[i],
            tat[i]);
    }

    return 0;
}

```

^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute
 ^X Exit ^R Read File ^N Replace ^U Paste ^J Justify

```
P2 34 56 2 36
hemasai@Hemasai:~/OSSP_Assignment/Scheduling$ nano round_robin.c
hemasai@Hemasai:~/OSSP_Assignment/Scheduling$ gcc round_robin.c -o round_robin
hemasai@Hemasai:~/OSSP_Assignment/Scheduling$ ./round_robin
Enter number of processes: 2
Enter burst time:
P1: 2
P2: 3
Enter time quantum: 4

Process BT    WT    TAT
P1      2      0      2
P2      3      2      5
hemasai@Hemasai:~/OSSP_Assignment/Scheduling$ |
```